LEMON: Factor-Level Semantic Alignment for Graph–Text Pairs

Anton Tomilov[1], Daria Gineva[2], Danil Tirskikh[2], Olesia Koroteeva[2], and Yuri Matveev[2]

[1] STC-Innovation, Saint Petersburg, Russian Federation
[2] ITMO University, Saint Petersburg, 197101, Russian Federation

Abstract. Graph–text evaluation often misses a central failure mode: a sentence may preserve the entity names while changing or omitting the relation encoded by the graph edge. We introduce LEMON (Linguistically Enriched Measure of Ontological Normalization), a diagnostic metric for bidirectional knowledge-graph (KG)–text fidelity. LEMON decomposes each graph predicate into weighted semantic factors and evaluates both KG→Text coverage and Text→KG recoverability. Its goal is not to be the most severe scalar detector. Instead, it explains relation-level errors by localizing them to argument roles, relation class, direction, polarity, or textual evidence. On WebNLG, Forward LEMON reaches 0.792 and Reverse LEMON reaches 0.779. A MINE-style comparison confirms that node information is easier to recover than edge information, while worked examples and controlled perturbations show why a factor trace is more informative than a single edge-failure label. The same inventory format is applied to DrugProt and BC5CDR, where biomedical predicates make role, direction, polarity, and causality explicit. A pilot review with four expert reviewers yields 85.0% exact agreement between LEMON and expert factor labels, and LEMON matches the expert majority on 27/28 rows with a $\geq 3/4$ majority. Code and reproducibility materials are available at `https://github.com/Cogniscy/lemon`.

Keywords: Graph-to-text evaluation · Semantic alignment · Knowledge graphs · WebNLG · Biomedical relation extraction · Predicate decomposition

1 Introduction

A graph edge is small, but it carries more than two names. In an edge (s, p, o), the predicate p fixes the relation between the arguments: who did what, to whom, in which direction, and with which semantic commitment. If a graph-to-text system mentions the entities but loses the predicate, the sentence may still look fluent while no longer saying the same thing. The reverse problem is similar: a reconstructed graph is useful only if it recovers the relation, not merely the endpoints.

This failure is easy to overlook. A sentence can mention *Alan Bean* and *Wheeler, Texas* without stating that Wheeler is his place of birth. A biomedical

abstract can preserve the same chemical and protein names while turning inhibition into activation. Standard WebNLG evaluation uses reference metrics such as BLEU, METEOR, chrF++, TER, BERTScore, and BLEURT, together with human criteria such as data coverage, relevance, correctness, and fluency [1–3]. These signals remain useful, but they rarely identify which part of a predicate was preserved, weakened, or lost.

LEMON treats the predicate as the unit that must be audited. Each predicate is decomposed into typed, weighted semantic factors. For example, *birthPlace* contains a person-like subject, a biographical relation, and a place-like object. A DrugProt inhibition predicate contains a chemical actor, a gene/protein target, negative regulation, directionality, and evidence form. The score asks whether these factors are expressed in text or recovered in a reconstructed graph, and records which factors are missing.

The setting is deliberately restricted. We do not attempt general text–text similarity or open-ended semantic parsing. The source graph supplies the initial semantic structure, and LEMON audits the predicate layers that should remain stable across graph and text. In this sense, the method is a controlled graph–text instance of a broader layered-semantics idea: represent relation meaning explicitly, then compare it through interpretable parts rather than through a single opaque similarity score.

The framework is bidirectional. In KG→Text, LEMON evaluates whether the text covers the semantic factors of the input graph. In Text→KG, it evaluates whether a reconstructed graph preserves the source predicate factors. We also implement a MINE-style node/edge baseline inspired by KGGen's Measure of Information in Nodes and Edges (MINE), which evaluates text-to-KG information retention through recovered nodes and relations [22]. This baseline is a comparison point, not a reproduction of the full KGGen benchmark.

The experiments follow one line of reasoning: factor coverage beyond entity overlap on WebNLG, reuse of the same factors in the reverse direction, controlled perturbations for intended semantic damage, transfer of the inventory format to DrugProt and BC5CDR, a pilot four-expert trace review, and a small three-judge LLM probe for residual recoverability.

The contributions are a bidirectional factor-level formulation for graph–text semantic alignment; fixed factor inventories for WebNLG, DrugProt, and BC5CDR; reproducible forward, reverse, perturbation, and ablation analyses; a MINE-style node/edge comparison; a pilot expert trace review; and an exploratory LLM reliability probe. The paper is diagnostic rather than leaderboard-oriented: it shows when entity, triple, node/edge, vector-space, and factor-level views disagree, and what those disagreements reveal about relation-level semantic preservation.

2 Related Work

2.1 RDF-to-text evaluation and reference metrics

The WebNLG Challenge made Resource Description Framework (RDF)-to-text generation a standard test bed: systems receive DBpedia-derived triples and

produce natural-language verbalizations [1]. WebNLG+ later added bilingual and bidirectional settings, so the same data family can support graph-to-text generation and text-to-graph recovery [2]. GEM placed such tasks within a broader natural language generation (NLG) evaluation suite, and recent LLM-based graph-to-text work, such as PlanGTG, revisits planning and attribution in graph-conditioned generation [3, 4].

These benchmarks often report sentence-level reference metrics. BLEU measures word n-gram precision; METEOR aligns words, stems, and synonyms while balancing precision and recall; chrF++ uses character n-gram F-score with a word n-gram extension; TER counts edits needed to transform an output into a reference [5–8]. BERTScore compares contextual token embeddings, and BLEURT learns a reference-based quality signal from human-rated examples [9, 10]. These metrics are useful for reference similarity and generation quality. They do not identify whether a source predicate preserved its argument roles, direction, polarity, or evidence.

2.2 Source-aware factuality and graph metrics

Source-aware metrics move closer to the input structure. PARENT weights n-gram precision and recall by evidence in the source table and reference text; Data-QuestEval casts data-to-text coverage as question answering; AlignScore learns a general alignment function for factual consistency [11–13]. FactSpotter is the closest prior graph-to-text evaluation approach: it checks whether a triple (s, p, o) is present in a generated text and correlates with human judgments of correctness, coverage, and relevance [14]. LEMON differs by replacing a binary whole-triple decision with weighted predicate-factor diagnosis.

Text-driven factuality metrics take the opposite route. They decompose generated text into verifiable claims and check support against evidence, as in FActScore and VERISCORE [15, 16]. Semantic graph metrics compare structures under node alignment, as in Smatch, S^2match, and Smatch++ for Abstract Meaning Representation (AMR) parsing [17–19]. LEMON is source-driven: it decomposes the graph predicate first and evaluates the text against that decomposition.

2.3 MINE-style recovery and LLM judging

Text-to-knowledge-graph (text-to-KG) evaluation asks whether extracted graphs retain information from text. KGGen introduces a language-model-based extractor and MINE, the Measure of Information in Nodes and Edges, for node and edge retention [22]. Our MINE-style baseline adapts this node/edge split to WebNLG graph–text pairs. It is a useful coarse graph-fidelity signal, but it does not separate polarity errors, role swaps, and missing relation class; all can appear as edge-level failure.

LLM-based evaluation can recover meaning that lexical rules miss, but it must be separated from reproducible scoring. G-Eval and MT-Bench illustrate both the scale of LLM judging and its calibration risks [26, 27]. We use LLMs

only in an exploratory factor-judging probe; the main tables are rule-based and reproducible without API calls.

2.4 Behavioral metric calibration

A metric should react predictably to controlled changes. CheckList introduced invariance and directional tests for NLP models [23]. Perturbation CheckLists and synthetic stress tests apply the same idea to NLG evaluation [24, 25]. Our perturbations follow this logic: relation deletion, argument swap, polarity flip, and relation blur test whether a metric reacts to intended semantic damage.

3 Method

3.1 Problem setting

Let a graph–text pair be (G, T), where G is a set of directed edges $e = (s, p, o)$ and T is a verbalization. The forward task asks whether the text covers the information in the graph. The reverse task asks whether a graph reconstructed from the text, denoted $\hat{G}(T)$, preserves the information in the original graph. LEMON uses the same predicate-level semantic representation in both directions.

3.2 Predicate factorization

Each predicate p is mapped to a weighted decomposition:

$$D(p) = \{(f_i, r_i, w_i)\}_{i=1}^{n},$$

where f_i is a semantic factor, r_i is its role, and $w_i > 0$ is a salience weight. Roles cover predicate meaning, subject and object domains, value domains, and modifiers such as direction, polarity, causality, or evidence. The representation is coarser than a full ontology but richer than a single predicate label. A text that keeps both entities but loses the relation cue should therefore receive partial credit rather than full coverage.

3.3 Factor inventory construction and weights

The inventories are fixed before scoring. They are manually specified diagnostic resources built from predicate labels, argument roles, domain constraints, entity types, and lexical evidence cues. Weights are normalized salience priors: predicate-meaning factors usually receive the largest share; domain factors and modifiers receive smaller shares when they are part of the predicate meaning. These weights are auditable design choices rather than learned parameters; a pilot expert trace review in Section 6.3 checks whether the resulting factor labels agree with human judgments.

Figure 1 shows two worked examples using current inventory weights. The point is not that LEMON is always numerically higher than a coarse node/edge score, but that a coarse relation mismatch can be decomposed into different semantic causes.

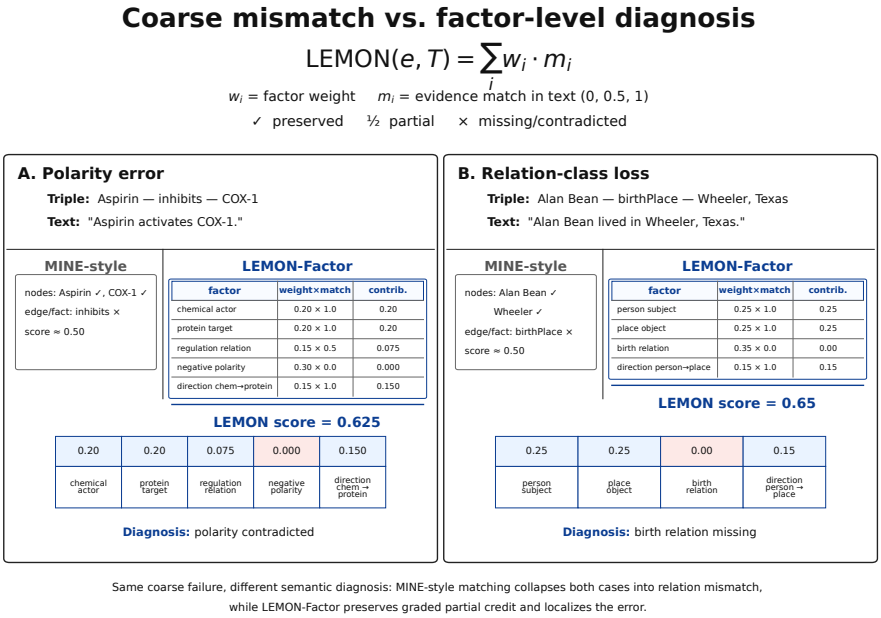

Fig. 1. Worked examples of factor-level scoring. The displayed weights come from the current predicate inventories. The MINE-style view collapses both cases into a coarse relation mismatch, whereas LEMON assigns partial credit to preserved factors and localizes the failure to polarity or relation class. The examples illustrate scoring mechanics, not a separate benchmark.

3.4 Forward LEMON: KG→Text coverage

For an edge e, text T, and component (f_i, r_i, w_i), an evidence function $E(f_i, r_i, e, T) \in [0, 1]$ estimates whether the component is expressed. In the deterministic implementation, entity and domain factors are supported by normalized label or token evidence, while relation factors are supported by predicate cues and factor cues. The edge-level forward score is:

$$S_F(e, T) = \frac{\sum_i w_i E(f_i, r_i, e, T)}{\sum_i w_i}.$$

Example-level and corpus-level scores are averages over edges with available decompositions. The implementation stores covered and missing factors for each edge, so the scalar score is accompanied by a diagnostic trace.

3.5 Reverse LEMON: Text→KG recoverability

The reverse direction first constructs a deterministic reconstructed graph $\hat{G}(T)$ from the text using entity-label evidence and predicate lexical cues. Reverse LEMON then compares each gold edge with its best recovered counterpart.

Subject and object recovery contribute through node/domain evidence; predicate recovery contributes through overlap between the gold predicate's factors and the recovered edge evidence. The reverse score therefore asks whether the source predicate's semantic factors survive the text-to-graph reconstruction.

The current reconstructed graph is a lexical pseudo-extractor, not a full KG extraction model. This choice makes the experiment reproducible and keeps the focus on metric behavior rather than extractor quality. Future work can replace $\hat{G}(T)$ with KGGen, GraphRAG-style extraction, OpenIE, or domain-specific extractors.

3.6 MINE-style node/edge baseline

To compare factor-level recoverability with node/edge information retention, we implement a MINE-style baseline inspired by KGGen's MINE. Each gold triple is treated as a fact, and the reconstructed graph is searched for matching or related nodes and edges. The deterministic score separates node information, edge information, and a composite partial-credit node/edge score. We report it as a local baseline rather than a reproduction of the original benchmark.

3.7 Controlled perturbation calibration

A metric that measures semantic alignment should degrade under meaning-destroying perturbations and remain comparatively stable under meaning-preserving perturbations. We therefore create deterministic perturbations of both text and reconstructed graphs. Text-side destructive perturbations include relation phrase deletion, object swaps, and predicate swaps. Graph-side destructive perturbations include node drops, edge drops, and predicate swaps. Preserving perturbations include alias-like rewrites and template paraphrases, which are flagged as clean or weak-template depending on their reliability. All perturbed files and operation-level manifests are saved, enabling audit of each injected change.

4 Data and Experimental Resources

4.1 WebNLG sample

The WebNLG experiments use a stratified English sample from the RDF-to-text benchmark introduced above. Each item pairs a small DBpedia-derived graph with one or more English verbalizations. The train split contains 200 examples and the development split contains 100 examples; both cover 16 WebNLG categories. The development set contains 308 graph edges. The sample is small by design: it keeps the scoring pipeline auditable while still exposing relation-level errors that entity overlap can hide.

4.2 Biomedical transfer resources

The biomedical transfer checks use two relation extraction corpora converted into the same graph–text JSONL format. DrugProt is a BioCreative VII resource for chemical–gene/protein relations with 13 relation types and 24,526 manually annotated relations [20]. BC5CDR is a BioCreative V collection of 1500 PubMed articles annotated for chemicals, diseases, and chemical-induced disease (CID) relations; it contains 3116 chemical–disease interactions [21]. In our processed subsets, DrugProt provides 500 train abstracts with 3588 edges, and BC5CDR provides 500 train and 500 development documents with 985 and 961 CID edges.

Biomedical text stresses the dimensions that motivate LEMON. In scientific abstracts, a change from inhibition to activation, a reversed chemical–protein relation, or a weakened causal claim can alter the interpretation of the result. Literal evidence such as dosages, measurements, and experimental conditions also becomes important when text is translated into a graph. DrugProt and BC5CDR are therefore used as diagnostic transfer settings: they test role, direction, polarity, causality, and evidence under domain conditions where surface entity overlap is not enough. We do not train or claim a new biomedical relation extraction system.

4.3 Predicate-factor inventories

The experiments use fixed JSON predicate-factor inventories. Each inventory contains a controlled factor vocabulary and one weighted decomposition per predicate. WebNLG uses 121 predicate decompositions and covers all 119 predicate labels observed in the evaluation files. DrugProt contains 13 biomedical relation decompositions and covers the 12 relation types observed in the processed subset. BC5CDR is intentionally a single-relation setting: its inventory decomposes the document-level chemical-induced disease relation into chemical exposure, disease outcome, causality, document-level evidence, and chemical-to-disease direction.

The inventories are rule-based diagnostic resources. Earlier LLM-generated decomposition candidates are kept as development material, but they are not treated as expert gold and are not used as the main evidence in the scoring tables below. The validator checks that coverage is complete for the current evaluation files and that each predicate decomposition has normalized weights.

5 Experiments

The experiments test one representation from several angles. WebNLG provides an explicit RDF-to-text setting. DrugProt and BC5CDR then test whether the same factor format remains meaningful when predicates encode biomedical roles, direction, polarity, and causality. Deterministic scoring uses JSONL inputs and fixed JSON factor inventories. LLM calls appear only in the final reliability probe; they are not used to construct the deterministic scores.

5.1 Forward and reverse factor scoring

The first WebNLG experiment scores development texts using predicate decompositions and lexical cues. We run an error-driven resource expansion: missing predicates are identified, deterministic decompositions are added, lexical cues are expanded, and coverage is recomputed. This separates missing evaluation resources from missing information in the text.

To test bidirectionality, we reconstruct a graph from each WebNLG text with a deterministic lexical pseudo-extractor and compute Reverse LEMON. The reverse score is compared with expanded Forward LEMON. A small forward–reverse gap would indicate that the same factor schema can support both KG→Text coverage and Text→KG recoverability.

5.2 Node/edge retention and perturbations

We compare Reverse LEMON with a MINE-style baseline inspired by KGGen's Measure of Information in Nodes and Edges. The deterministic variant reports node information, edge information, and a composite node/edge score. A locked 50-fact subset shows how binary fact recoverability differs from a partial-credit node/edge composite. These values are diagnostic baselines, not a reproduction of the original KGGen/MINE benchmark.

Controlled perturbations are then injected into text and reconstructed graphs. The operators keep the source graph fixed and change only the text, which lets us ask whether a metric reacts to intended semantic damage rather than to a change in the reference graph. The final scoring uses five operators: node deletion, relation-cue deletion, argument swap, polarity flip, and relation blur. Polarity flips are most informative for DrugProt and BC5CDR, where relation labels explicitly encode activation, inhibition, regulation, or causality.

We score the shared perturbation files with deterministic baselines: entity recall, label match, a binary triple-style match, a MINE-style node/edge harmonic score, a label-only LEMON variant, and the full deterministic LEMON proxy based on the fixed factor inventories and perturbation metadata. The same scoring layer is used for the compact ablation study. An offline character n-gram vector baseline is reported as a lexical/topical control, not as a dense embedding benchmark.

5.3 Pilot expert trace review

To test whether LEMON's factor traces match human judgments, we prepared a compact expert review form with eight diagnostic graph–text examples and 35 factor rows. Four completed expert submissions were included and one empty template was excluded. Reviewers assigned each factor one of the same ordinal preservation labels used by LEMON: preserved, mostly preserved, partial, mostly lost, lost, or contradicted. We then measured exact and weighted LEMON–expert agreement, chance-corrected agreement, and inter-expert reliability.

5.4 Exploratory LLM factor judging

The LLM reliability probe asks a different question: whether model judges can still recover the same factors after deterministic perturbations mark them as damaged. We sample perturbed instances from WebNLG, DrugProt, and BC5CDR and ask three independent LLM judges to label each semantic factor as covered, partially covered, or absent. The judges are Gemini 2.0 Flash, GPT-4o-mini, and Llama-3.1-70B. The prompts use JSON-schema constrained outputs through OpenRouter. The probe is reported as complementary diagnostic evidence.

6 Results

6.1 From entity overlap to relation factors

The first WebNLG experiment shows why predicate resources must be explicit. With the initial factor dictionary, only 140 of 308 development edges could be scored, and the missing decomposition rate was 0.546. After rule-based decomposition and cue expansion, all 308 edges had predicate factors and Forward LEMON rose from 0.368 to 0.792. This increase does not mean that all WebNLG texts are faithful. It shows that relation-level diagnosis requires predicate resources before text quality can be evaluated.

Lexical controls identify the added signal. Exact entity-label overlap is already high at 0.922 and does not change under predicate-resource expansion. Token cosine and Jaccard remain fixed because the graph–text strings do not change. LEMON changes because it checks relation roles, predicate meanings, and lexical evidence instead of counting entity mentions alone.

The reverse run checks the same representation from text back to graph. A lexical pseudo-extractor recovered 390 of 408 nodes and 196 of 308 edges from the WebNLG texts. Reverse LEMON reached 0.779, close to the expanded Forward score of 0.792. The MINE-style comparison explains the remaining difficulty: its composite score is 0.803, but the locked 50-fact subset has node information of 0.960 and edge/fact information of only 0.540. The relation is harder to recover than the names of the entities.

A relation-deletion sanity check isolates the same failure mode. Removing relation evidence changes exact entity-label coverage by only 0.002, while Forward LEMON drops by 0.211. Predicate cue coverage and MINE fact recoverability fall by 0.364 and 0.354. Preserving entity names is therefore not the same as preserving the predicate.

6.2 Cross-domain perturbation sensitivity

The cross-domain perturbation files test which metrics react when controlled edits damage relation meaning in WebNLG, DrugProt, and BC5CDR. The current inventories audit entity/domain support, predicate cues, participant roles,

Table 1. Perturbation sensitivity as mean score drop. Larger values indicate that a metric reacts more strongly to the perturbation; unchanged or small drops mark blind spots rather than missing computations.

Perturbation	Entity	Triple	Node/edge	Label-only	LEMON-Factor
Node deletion	0.326	0.675	0.657	0.323	0.352
Edge deletion	0.215	0.655	0.637	0.482	0.281
Argument swap	0.214	0.540	0.521	0.321	0.413
Polarity flip	0.215	0.546	0.526	0.337	0.414
Relation blur	0.216	0.568	0.549	0.385	0.281

Table 2. Ablation sensitivity. Mean drop is the average perturbation response. Gain is full drop minus ablated drop; positive values mean that removing the component reduces sensitivity.

Variant	Mean drop	Gain vs. full
LEMON-full	0.320	0.000
Label-only	0.417	-0.097
Unweighted	0.310	0.010
Minus roles	0.185	0.135
Minus direction	0.310	0.010
Minus polarity	0.304	0.016
Minus evidence	0.290	0.030

direction, polarity, and textual evidence. A perturbation can be diagnosed only when the corresponding factor group is present in the inventory.

Table 1 reports normalized score loss under five perturbations. The columns are different preservation signals rather than one shared accuracy scale: entity recall tracks entity survival; triple match and MINE-style node/edge scoring act as coarse edge/fact alarms; LEMON label-only removes factor decomposition; LEMON is the full weighted factor score.

The table should not be read as a global ranking. Triple match and MINE-style scoring often react more strongly because they treat edge damage as a coarse failure. LEMON is most useful when the reason for failure matters. It reacts to argument swaps and polarity flips when those dimensions are encoded, and it links the response to factor groups that can be inspected.

Table 2 is an internal component analysis, not a comparison with MINE-style or triple matching. Removing participant-role factors produces the largest positive gain for the full model over the ablated variant, so role factors are the largest observed source of diagnostic sensitivity. Evidence, polarity, and direction have smaller effects, mainly in biomedical predicates where these factors are explicit. Label-only scoring can produce a stronger scalar reaction when surface labels disappear, but it cannot say whether the damage concerns roles, direction, polarity, or evidence.

6.3 Pilot expert trace review

The expert review directly addresses whether LEMON's factor-level traces agree with human assessments of relation failure. Four expert reviewers completed the

Table 3. Exploratory LLM reliability probe. Judges are Gemini 2.0 Flash, GPT-4o-mini, and Llama-3.1-70B. Mean score treats covered, partial, and absent factor judgments as 1, 0.5, and 0; agreements are computed only on overlapping factor judgments.

Probe	Items	Judges	Mean score	Pairwise agr.	Det. agr.
LLM factor judging	45	3	0.809	0.597	0.561

form, yielding 140 expert–factor judgments over 35 factor rows. LEMON reached 85.0% exact agreement with expert labels (119/140; Wilson 95% CI: 78.2–90.0%) and 85.8% weighted exact agreement. Chance-corrected agreement was substantial: pooled Cohen's $\kappa = 0.791$, Fleiss' inter-expert $\kappa = 0.689$, and Krippendorff's ordinal $\alpha = 0.871$. Experts formed a $\geq 3/4$ majority on 28 rows; LEMON matched that majority on 27 rows (96.4%), and every LEMON label was supported by at least one expert. Disagreements concentrated in relation-blur and relation-missing cases, especially predicate-meaning, direction, and evidence factors.

6.4 LLM factor judging as a complementary view

The LLM probe evaluates semantic recoverability rather than rule-based perturbation labels. Each judge saw the source edge, the factor inventory, and the perturbed text, then labeled each factor as covered, partial, or absent. Table 3 summarizes Gemini 2.0 Flash, GPT-4o-mini, and Llama-3.1-70B.

The mean LLM factor score is 0.809, pairwise agreement is 0.597, and agreement with rule-based perturbation labels is 0.561. These values indicate moderate model-judge stability. Rule-based perturbations record intended damage; LLM judges estimate how much of the original factor remains recoverable from context. The expert review above tests direct agreement with human factor labels, while model-based judging shows which failures remain inferable.

7 Limitations and Error Sources

Two error sources recur in the experiments. The first is inventory incompleteness: predicates without decompositions, or with weak lexical cues, cannot be scored reliably. The error-driven expansion step exposes this issue before it contaminates the metric, but fallback decompositions remain less informative than expert-designed ones. The second is relation recoverability. WebNLG texts often repeat entity labels, so node recovery is high, while edge and fact recovery are substantially lower. The factor trace is the practical output in such cases: a low score can be traced to a predicate, a factor role, a missing cue, or weak subject/object evidence.

The expert review is preliminary rather than a final gold-standard validation. It covers 35 factor rows and four completed submissions, so it is best read as a pilot trace-review check rather than a large blind annotation campaign. The factor inventories remain rule-based resources built from predicate labels, schema constraints, domain roles, lexical cues, and development-time candidates. Prior

LLM-as-judge work also shows that model judges can scale evaluation while introducing preference, calibration, and reasoning biases; therefore the LLM probe is kept separate from the human trace review [26, 27].

The evidence layer is limited. It can find labels and known cues, but it can miss implicit relations, long-range entailment, and paraphrases outside the cue dictionary. A stronger implementation should add embedding, entailment, or QA-based evidence while preserving the factor trace.

The reverse pipeline uses a lexical pseudo-extractor rather than a full text-to-KG system. This keeps the experiment controlled, but the reverse scores should be read as metric-behavior diagnostics. Similarly, the MINE-style baseline adapts KGGen's node/edge information idea to graph–text pairs; it is not a reproduction of the original MINE benchmark.

LEMON also has a narrower scope than general semantic similarity. It assumes that a source graph, relation annotation, or predicate inventory is available. The present inventories cover roles, direction, polarity, causality, and evidence where those dimensions are encoded by the predicate resources. They do not model pragmatic phenomena such as irony, implicature, presupposition, stance, or rhetorical force. The perturbations are auditable but template-like; they support construct-validity checks for relation deletion, argument swap, and polarity flip, but they do not prove robustness to natural paraphrase or state-of-the-art biomedical extraction performance.

8 Conclusion

LEMON evaluates KG–text fidelity by opening graph predicates into weighted semantic factors. It does not replace sentence-level metrics, triple matching, or MINE-style node/edge scoring. Its role is diagnostic: when a relation fails, it identifies whether the failure concerns relation class, roles, direction, polarity, or evidence.

The experiments support this distinction. On WebNLG, the same factor representation supports forward coverage and reverse recoverability. The MINE-style comparison shows that nodes are easier to recover than edges, and the relation-deletion check shows that entity-label coverage can remain almost unchanged while relation coverage drops. Perturbation tests show that LEMON reacts to the semantic dimensions represented in each inventory, with biomedical predicates making polarity, direction, causality, and evidence especially visible.

The component, expert-review, and LLM analyses clarify the result. Role factors contribute most to diagnostic sensitivity under controlled damage. Label-only scoring can be more severe as a scalar detector, but it cannot explain whether the damage concerns role assignment, direction, polarity, or evidence. The expert trace review shows substantial agreement between LEMON and human factor labels, while LLM judges provide a complementary view of residual recoverability.

The main contribution is a layer between whole-triple factuality and sentence-level similarity: LEMON gives a reproducible factor trace for graph–text errors

that coarser scores collapse into one number. Future work should scale the expert review to larger blind annotation sets, strengthen the evidence layer with paraphrase and entailment signals, and repeat the reverse experiment with stronger text-to-KG extractors.

References

1. Gardent, C., Shimorina, A., Narayan, S., Perez-Beltrachini, L.: The WebNLG Challenge: Generating Text from RDF Data. In: INLG, pp. 124–133 (2017). https://doi.org/10.18653/v1/W17-3518
2. Castro Ferreira, T., et al.: The 2020 Bilingual, Bi-Directional WebNLG+ Shared Task: Overview and Evaluation Results. In: Proceedings of the 3rd International Workshop on Natural Language Generation from the Semantic Web (2020)
3. Gehrmann, S., Adewumi, T., Aggarwal, K., et al.: The GEM Benchmark: Natural Language Generation, its Evaluation and Metrics. In: Proceedings of the 1st Workshop on Natural Language Generation, Evaluation, and Metrics, pp. 96–120 (2021). https://doi.org/10.18653/v1/2021.gem-1.10
4. He, J., Yang, Y., Long, W., Xiong, D., Gutierrez Basulto, V., Pan, J.Z.: Evaluating and Improving Graph to Text Generation with Large Language Models. In: NAACL-HLT, pp. 10219–10244 (2025). https://doi.org/10.18653/v1/2025.naacl-long.513
5. Papineni, K., Roukos, S., Ward, T., Zhu, W.-J.: BLEU: A Method for Automatic Evaluation of Machine Translation. In: ACL, pp. 311–318 (2002)
6. Banerjee, S., Lavie, A.: METEOR: An Automatic Metric for MT Evaluation with Improved Correlation with Human Judgments. In: ACL Workshop on Intrinsic and Extrinsic Evaluation Measures for Machine Translation and/or Summarization, pp. 65–72 (2005)
7. Popovic, M.: chrF++: Words Helping Character N-grams. In: Proceedings of the Second Conference on Machine Translation, pp. 612–618 (2017)
8. Snover, M., Dorr, B., Schwartz, R., Micciulla, L., Makhoul, J.: A Study of Translation Edit Rate with Targeted Human Annotation. In: AMTA, pp. 223–231 (2006)
9. Zhang, T., Kishore, V., Wu, F., Weinberger, K.Q., Artzi, Y.: BERTScore: Evaluating Text Generation with BERT. In: ICLR (2020)
10. Sellam, T., Das, D., Parikh, A.P.: BLEURT: Learning Robust Metrics for Text Generation. In: ACL, pp. 7881–7892 (2020)
11. Dhingra, B., Faruqui, M., Parikh, A., Chang, M.-W., Das, D., Cohen, W.W.: Handling Divergent Reference Texts when Evaluating Table-to-Text Generation. In: ACL, pp. 4884–4895 (2019). https://doi.org/10.18653/v1/P19-1483
12. Rebuffel, C., Scialom, T., Soulier, L., Piwowarski, B., Lamprier, S., Staiano, J., Scoutheeten, G., Gallinari, P.: Data-QuestEval: A Referenceless Metric for Data-to-Text Semantic Evaluation. In: EMNLP, pp. 8029–8036 (2021). https://doi.org/10.18653/v1/2021.emnlp-main.633
13. Zha, Y., Yang, Y., Li, R., Hu, Z.: AlignScore: Evaluating Factual Consistency with a Unified Alignment Function. In: ACL, pp. 11328–11348 (2023). https://doi.org/10.18653/v1/2023.acl-long.634
14. Zhang, K., Balalau, O., Manolescu, I.: FactSpotter: Evaluating the Factual Faithfulness of Graph-to-Text Generation. In: Findings of EMNLP, pp. 10025–10042 (2023). https://doi.org/10.18653/v1/2023.findings-emnlp.672

15. Min, S., Krishna, K., Lyu, X., Lewis, M., Yih, W.-T., Koh, P.W., Iyyer, M., Zettlemoyer, L., Hajishirzi, H.: FActScore: Fine-Grained Atomic Evaluation of Factual Precision in Long Form Text Generation. In: EMNLP, pp. 12076–12100 (2023). https://doi.org/10.18653/v1/2023.emnlp-main.741
16. Sun, H., et al.: VERISCORE: Evaluating the Factuality of Verifiable Claims in Long-Form Text Generation. arXiv:2406.19276 (2024)
17. Cai, S., Knight, K.: Smatch: An Evaluation Metric for Semantic Feature Structures. In: ACL (2013)
18. Opitz, J., Daza, A., Frank, A.: AMR Similarity Metrics from Principles. Transactions of the Association for Computational Linguistics 9, 522–538 (2021). https://doi.org/10.1162/tacl_a_00329
19. Opitz, J.: Smatch++: Standardized and Extended Evaluation of Semantic Graphs. In: Findings of EACL, pp. 1595–1608 (2023). https://doi.org/10.18653/v1/2023.findings-eacl.118
20. Miranda-Escalada, A., Mehryary, F., Luoma, J., et al.: Overview of DrugProt BioCreative VII track: quality evaluation and large scale text mining of drug-gene/protein relations. Database 2023, baad080 (2023). https://doi.org/10.1093/database/baad080
21. Li, J., Sun, Y., Johnson, R.J., et al.: BioCreative V CDR task corpus: a resource for chemical disease relation extraction. Database 2016, baw068 (2016). https://doi.org/10.1093/database/baw068
22. Mo, B., Yu, K., Kazdan, J., Cabezas, J., Mpala, P., Yu, L., Cundy, C., Kanatsoulis, C., Koyejo, S.: KGGen: Extracting Knowledge Graphs from Plain Text with Language Models. arXiv:2502.09956 (2025)
23. Ribeiro, M.T., Wu, T., Guestrin, C., Singh, S.: Beyond Accuracy: Behavioral Testing of NLP Models with CheckList. In: ACL, pp. 4902–4912 (2020). https://doi.org/10.18653/v1/2020.acl-main.442
24. Sai, A.B., Dixit, T., Sheth, D.Y., Mohan, S., Khapra, M.M.: Perturbation CheckLists for Evaluating NLG Evaluation Metrics. In: EMNLP, pp. 7282–7296 (2021). https://doi.org/10.18653/v1/2021.emnlp-main.575
25. He, T., Zhang, J., Wang, T., Kumar, S., Cho, K., Glass, J.: On the Blind Spots of Model-Based Evaluation Metrics for Text Generation. In: ACL, pp. 12067–12097 (2023). https://doi.org/10.18653/v1/2023.acl-long.674
26. Liu, Y., Iter, D., Xu, Y., Wang, S., Xu, R., Zhu, C.: G-Eval: NLG Evaluation using GPT-4 with Better Human Alignment. In: EMNLP, pp. 2511–2522 (2023). https://doi.org/10.18653/v1/2023.emnlp-main.153
27. Zheng, L., Chiang, W.-L., Sheng, Y., et al.: Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. In: NeurIPS (2023)